

Т6. Случайная величина имеет распределение Парето:

$$p(x) = \begin{cases} \frac{\theta-1}{x^\theta}, & x \geq 1 \\ 0, & x < 1 \end{cases}, \quad \theta > 1$$

$$a) L(\theta) = \prod_{i=1}^n p(x_i, \theta) = (\theta-1)^n \cdot \frac{1}{\prod_{i=1}^n x_i^\theta} \{x_i \geq 1\}$$

$$\begin{aligned} \ln L(\theta) &= \ln(\theta-1)^n + \ln\left(\prod_{i=1}^n x_i\right)^{-\theta} = \\ &= n \ln(\theta-1) - \theta \ln\left(\prod_{i=1}^n x_i\right); \quad (\ln L(\theta))'_\theta = \\ &= \frac{n}{\theta-1} - \ln\left(\prod_{i=1}^n x_i\right) = 0 \quad \theta-1 = \frac{n}{\sum_{i=1}^n \ln x_i} \Rightarrow \end{aligned}$$

$$\Rightarrow \theta = 1 + \frac{n}{\sum_{i=1}^n \ln x_i}; \quad (\ln L(\theta))''_\theta = -\frac{n}{(\theta-1)^2} < 0$$

$$\Rightarrow \max \Rightarrow \left[\hat{\theta} = 1 + \frac{n}{\sum_{i=1}^n \ln x_i} \right]$$

$$b) p_{S(k)} = n C_{n-1}^{k-1} F^{k-1} (1-F)^{n-k} \cdot p(t) \quad (\text{as } T_2)$$

$$F(x) = \int_{-\infty}^x p(t) dt = \begin{cases} 0, & x \leq 1 \\ \int_1^x \frac{\theta-1}{t^\theta} dt, & x > 1 \end{cases}$$

$$= \begin{cases} 0, & x \leq 1 \\ \theta-1 \int_1^x \frac{dt}{t^\theta} = (\theta-1) \frac{t^{-\theta+1}}{-\theta+1} \Big|_1^x = \theta-1 \left(\frac{1}{(1-\theta)x^{-1+\theta}} - \frac{1}{1-\theta} \right), & x > 1 \end{cases}$$

$$= \begin{cases} 0, & x \leq 1 \\ -\frac{1}{x^{-1+\theta}+1} = 1 - \frac{1}{x^{-1+\theta}}, & x > 1 \end{cases}$$

Медиана: $F(X_{1/2}) = 1/2 \Rightarrow 1 - \frac{1}{X_{1/2}^{\theta-1}} = 1/2$

$\Rightarrow X_{1/2}^{\theta-1} = 2 \quad X_{1/2} = \sqrt[\theta-1]{2}$

$(\theta-1) \ln X_{1/2} = \ln 2$

Пачи. случ. вел $y = \ln \xi$. $\Phi(y) = P(y \leq y) =$
 $= P(\ln \xi \leq y) = \int_{\ln x \leq y} dF(x) = \int_{-\infty}^{\infty} dF(x) =$



$= F(e^y - 0) - 0 = F(e^y) =$

$= \begin{cases} 0, & e^y \leq 1 \\ 1 - \frac{1}{e^{y(\theta-1)}}, & e^y > 1 \end{cases}$

$= \begin{cases} 0, & y \leq 0 \\ 1 - e^{-y(\theta-1)}, & y > 0 \end{cases}; \quad \phi(y) = \Phi' = \begin{cases} 0, & y \leq 0 \\ (\theta-1)e^{-y(\theta-1)}, & y > 0 \end{cases}$

$\Rightarrow y_k \sim \Gamma(\theta-1, 1) \left(\frac{(\theta-1)^{\theta-1}}{\Gamma(\theta-1)} x^{\theta-2} e^{-x} \right)$
 $= (\theta-1) e^{-x(\theta-1)} \{ (0; +\infty) \}; \quad \Gamma(1) = \int_0^{\infty} e^{-x} dx =$
 $= -e^{-x} \Big|_0^{\infty} = 1$

Покажем, что y_k - незав. П.к. ξ_k - незав.
 как э-ты выборки, то:

$\Phi(y_1, \dots, y_k) = P(y_1 \leq y_1, \dots, y_k \leq y_k) = P(\ln \xi_1 \leq y_1, \dots,$
 $\ln \xi_k \leq y_k) = \int_{\ln x_1 \leq y_1} \dots \int_{\ln x_k \leq y_k} dF(x_1, \dots, x_k) = \int_{\ln x_1 \leq y_1} \dots \int_{\ln x_k \leq y_k} p(x_1, \dots, x_k) dx =$

$\stackrel{\text{п.к. незав.}}{=} \int_{\ln x_1 \leq y_1} \dots \int_{\ln x_k \leq y_k} p(x_1) \dots p(x_k) dx_1 \dots dx_k \quad \Leftrightarrow$

$$\Rightarrow \left[\begin{matrix} 0_{y_i} \\ \theta_{y_i} \end{matrix} \right] y_i : y_i \stackrel{\theta_{y_i}}{\leq} 0$$

$$\int_{-\infty}^{\infty} p(x_1) dx_1 \int_{-\infty}^{\infty} p(x_2) dx_2 \dots \int_{-\infty}^{\infty} p(x_k) dx_k = \varphi_{y_1}(y_1) \dots \varphi_{y_k}(y_k) \Rightarrow \text{независим.}$$

$$\Rightarrow \text{по об-б-е суммы: } S = y_1 + \dots + y_n = S =$$

$$= \sum_{i=1}^n \ln x_i \sim \Gamma(\theta-1, n)$$

$$P(x) = \frac{(\theta-1)^n}{\Gamma(n)} x^{n-1} e^{-(\theta-1)x} \quad \{ (0, +\infty) \}$$

$$\text{Полн. } 2(1+\theta)S : \hat{F} = P(2(1+\theta)S < y) =$$

$$= P\left(S < \frac{y}{2(1+\theta)}\right) = F\left(\frac{y}{2(1+\theta)}\right) =$$

$$= \int_{-\infty}^{\frac{y}{2(1+\theta)}} p_{\Gamma(\theta-1, n)}(x) dx = \int_{-\infty}^{\frac{y}{2(1+\theta)}} \frac{(\theta-1)^n}{\Gamma(n)} x^{n-1} e^{-(\theta-1)x} dx =$$

$$= \begin{cases} -2(1-\theta)x = t \\ -2(1-\theta)dx = dt \\ x = \frac{t}{2(1-\theta)} \end{cases} = \int_0^{\frac{y}{2(1+\theta)}} \frac{(\theta-1)^n}{\Gamma(n)} \frac{t^{n-1}}{2^{n-1}(\theta-1)^{n-1}} e^{-\frac{(\theta-1)t}{2(1-\theta)}} \frac{dt}{2(1-\theta)} =$$

$$= \int_0^{\frac{y}{2(1+\theta)}} \frac{t^{n-1}}{2^n \Gamma(n)} e^{-t/2} dt \Rightarrow \tilde{p}(x) = \tilde{F}' = \frac{x^{n-1}}{2^n \Gamma(n)} e^{-x/2} \quad \{ (0, +\infty) \}$$

$$\Rightarrow 2(\theta-1)S \sim \chi^2(2n)$$

$$\text{Планим образовать } f(\vec{X}_n, x_{1/2}) = 2 \frac{\ln 2}{\ln x_{1/2}} S =$$

$$= \frac{2 \ln 2}{\ln x_{1/2}} \sum_{i=1}^n \ln x_i \sim \chi^2(2n)$$

$$t_1 = \chi_{\frac{1-\beta}{2}}^2(2n), \quad t_2 = \chi_{\frac{1+\beta}{2}}^2(2n)$$

$$P\left(t_1 < \frac{2 \ln 2}{\ln x_{1/2}} S < t_2\right) = \beta$$

$$\Rightarrow P\left(\frac{2 \ln 2 \cdot S}{t_2} < \ln X_{1/2} < \frac{2 \ln 2 \cdot S}{t_1}\right) = \sqrt{3}$$

$$\left[e^{\frac{2 \ln 2 \cdot S}{t_2}} < X_{1/2} < e^{\frac{2 \ln 2 \cdot S}{t_1}} \right] \quad - \text{ доверительный интервал для медианы}$$

$$\left(e^{\ln 2} \right)^{\frac{2S}{t}}$$

$$\Downarrow$$

$$\underline{2^{\frac{2S}{t_2}} < X_{1/2} < 2^{\frac{2S}{t_1}}}$$

c) $\Theta = (1; +\infty)$ — открытое ин-во.

Проверка сильной регулярности по Θ :

1) $F(x, \Theta) = \begin{cases} 0 & x \leq 1 \\ 1 - \frac{1}{x^{\Theta-1}} & x > 1 \end{cases}$ — 2-й непер. дифф. по Θ на $\Theta > 1$ $\oplus$

2) Регулярность по Θ :

$F(x, \Theta)$ — непер. дифф. по Θ на $\Theta > 1$,

$$\int_1^{+\infty} \frac{\partial F}{\partial \Theta} dx = \int_1^{+\infty} \frac{x^\Theta - (\Theta-1)x^\Theta \ln x}{x^{2\Theta}} dx =$$

$$= \int_1^{+\infty} \left(\frac{1}{x^\Theta} - \frac{(\Theta-1) \ln x}{x^\Theta} \right) dx = I_1 - I_2 = \frac{1}{\Theta-1} - \frac{1}{\Theta-1} = 0 \oplus$$

$$\left\{ \begin{aligned} I_1 &= \int_1^{+\infty} \frac{dx}{x^\Theta} = \frac{x^{-\Theta+1}}{-\Theta+1} \Big|_1^{+\infty} = \frac{1}{(1-\Theta)x^{\Theta-1}} \Big|_1^{+\infty} = \left\{ \begin{matrix} \Theta > 1 \\ \Theta-1 > 0 \end{matrix} \right\} = \frac{1}{\Theta-1} \\ I_2 &= \int_1^{+\infty} \frac{(\Theta-1) \ln x}{x^\Theta} dx = \int_1^{+\infty} (\Theta-1) \ln x d \frac{x^{1-\Theta}}{1-\Theta} = \\ &= (\Theta-1) \left[\frac{\ln x}{(1-\Theta)x^{\Theta-1}} \Big|_1^{+\infty} - \int_1^{+\infty} \frac{1}{(1-\Theta)x^{\Theta-1}} \cdot \frac{1}{x} dx \right] = \end{aligned} \right\}$$

$$\textcircled{=} (\theta-1) \left[-\lim_{x \rightarrow +\infty} \frac{1/x}{(\theta-1)^2 x^{\theta-2}} - 0 + \frac{1}{\theta-1} \int_1^{+\infty} \frac{1}{x^\theta} dx \right] =$$

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$$= \frac{\theta-1}{\theta-1} \int_1^{+\infty} \frac{1}{x^\theta} dx = \frac{1}{\theta-1}$$

$$I(\theta) = M \left[\left(\frac{\partial \ln p(x, \theta)}{\partial \theta} \right)^2 \right] \textcircled{=}$$

$$\begin{cases} p(x, \theta) = \frac{\theta-1}{x^\theta} \{1; +\infty\}; \ln p(x, \theta) = \ln(\theta-1) - \theta \ln x \\ \frac{\partial \ln p}{\partial \theta} = \frac{1}{\theta-1} - \ln x, \left(\frac{\partial \ln p}{\partial \theta} \right)^2 = \frac{1}{(\theta-1)^2} - \frac{2 \ln x}{\theta-1} + (\ln x)^2 \end{cases}$$

$$\textcircled{=} \int_1^{+\infty} \left(\frac{1}{(\theta-1)^2} - \frac{2 \ln x}{\theta-1} + (\ln x)^2 \right) \frac{\theta-1}{x^\theta} dx =$$

$$= \int_1^{+\infty} \frac{1}{(\theta-1) x^\theta} dx - \int_1^{+\infty} \frac{2 \ln x}{x^\theta} dx + \int_1^{+\infty} \frac{(\theta-1) \ln^2 x}{x^\theta} dx =$$

$$= I_1 - 2I_2 + (\theta-1) I_3 \textcircled{=}$$

$$\left\{ I_1 = \int_1^{+\infty} \frac{1}{(\theta-1) x^\theta} dx = \frac{1}{(\theta-1)^2}; \quad I_2 = \int_1^{+\infty} \frac{\ln x}{x^\theta} dx = \frac{1}{(\theta-1)^2} \right\}$$

$$I_3 = \int_1^{+\infty} \frac{\ln^2 x}{x^\theta} dx = \int_1^{+\infty} \ln^2 x d \left(\frac{x^{1-\theta}}{1-\theta} \right) =$$

$$= \frac{x^{1-\theta}}{1-\theta} \ln^2 x \Big|_1^{+\infty} - \int_1^{+\infty} \frac{1}{(1-\theta) x^{\theta-1}} \cdot \frac{2 \ln x}{x} dx =$$

$$= \frac{2}{\theta-1} \int_1^{+\infty} \frac{\ln x}{x^\theta} dx = \frac{2}{(\theta-1)^3}$$

$$\textcircled{=} \frac{1}{(\theta-1)^2} - \frac{2}{(\theta-1)^2} + \frac{2}{(\theta-1)^2} = \frac{1}{(\theta-1)^2} \quad \text{непрер. и} \\ \text{полным. по } \theta \geq 1$$

$\Rightarrow$ модель регулярна $\oplus$

$$3) \frac{\partial^2}{\partial \theta^2} \int_{-\infty}^{+\infty} dF(x, \theta) \stackrel{?}{=} \int_{-\infty}^{+\infty} d \frac{\partial^2 F(x, \theta)}{\partial \theta^2}$$

$$\frac{\partial^2}{\partial \theta^2} \int_{-\infty}^{+\infty} dF(x, \theta) = \frac{\partial^2}{\partial \theta^2} (1) = 0$$

$$\frac{\partial F}{\partial \theta} = \frac{\partial}{\partial \theta} \left(1 - \frac{1}{x^{\theta-1}} \right) = \frac{\partial}{\partial \theta} (1 - x^{1-\theta}) =$$

$$= + x^{1-\theta} \ln x, \quad \frac{\partial^2 F}{\partial \theta^2} = - x^{1-\theta} \ln^2 x$$

$$d \left(\frac{\partial^2 F}{\partial \theta^2} \right) = - \left((1-\theta) x^{-\theta} \ln^2 x + x^{1-\theta} \frac{2 \ln x}{x} \right) dx =$$

$$= - \left(\frac{(\theta-1) \ln^2 x}{x^\theta} - \frac{2 \ln x}{x^\theta} \right) dx; \quad \left\{ \frac{\partial^2 F}{\partial \theta^2} = - x^{1-\theta} \ln^2 x \right\}_{(1; +\infty)}$$

$$\int_{-\infty}^{+\infty} = \int_1^{+\infty} \left(\frac{(\theta-1) \ln^2 x}{x^\theta} - \frac{2 \ln x}{x^\theta} \right) dx = \frac{2}{(\theta-1)^2} - \frac{2}{(\theta-1)^2} = 0 \oplus$$

$\Rightarrow$ Модель явл. только регулярной.

$$g(\theta) = \theta, \quad \nabla g = 1, \quad I(\theta) = \frac{1}{(\theta-1)^2}$$

$$\frac{\sqrt{n}(\bar{\theta} - \theta)}{\sigma(\bar{\theta})} \rightsquigarrow N(0, 1), \quad \sigma^2(\theta) =$$

$$= \nabla g^T I(\theta) \nabla g = (\theta-1)^2 \Rightarrow \sigma = \theta-1$$

$$\sqrt{n} \frac{\bar{\theta} - \theta}{\bar{\theta} - 1} \rightsquigarrow N(0, 1), \quad \bar{\theta} = 1 + \frac{n}{\sum \ln x_i}$$

$$t_1 = u_{\frac{1-\beta}{2}}, \quad t_2 = u_{\frac{1+\beta}{2}} - \text{квантили}$$

$$P\left(z_1 \leq \sqrt{n} \frac{\hat{\theta} - \theta}{\hat{\theta} - 1} \leq z_2\right) = \beta$$

$$\frac{z_1(\hat{\theta} - 1)}{\sqrt{n}} < \hat{\theta} - \theta < \frac{z_2(\hat{\theta} - 1)}{\sqrt{n}}$$

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$$\left[\hat{\theta} - \frac{z_2(\hat{\theta} - 1)}{\sqrt{n}} < \theta < \hat{\theta} - \frac{z_1(\hat{\theta} - 1)}{\sqrt{n}} \right]$$

симм.
довер.
интервал

Сравнение интервалов:

$\theta = 2$

Доверительный интервал для медианы : 1.784301 < median < 2.357835, длина = 0.573534

Непараметрический бутстрап для медианы : 1.689177 < median < 2.268982, длина = 0.579805

Параметрический бутстрап для медианы : 1.698626 < median < 2.255045, длина = 0.556420

Асимптотический доверительный интервал для параметра θ : 1.798534 < θ < 2.187861, длина = 0.389326

Непараметрический бутстрап для параметра θ : 1.743338 < θ < 2.166901, длина = 0.423563

Параметрический бутстрап для параметра θ : 1.764372 < θ < 2.160350, длина = 0.395978